

Design and Implementation of a Generic Monte Carlo Pricer in Python

Jérôme Lelong

Grenoble INP - Ensimag, LJK

April 15, 2026

Plan

- 1 Objectives
- 2 The multi-dimensional Black-Scholes model
- 3 The discrete time portfolio
- 4 In practice

Objectives (1)

- ▶ Price and hedge path dependent options in a multi-dimensional Black-Scholes model.
- ▶ We consider options with payoffs of the form

$$\varphi(S_{t_0}, S_{t_1}, \dots, S_{t_N})$$

where $0 = t_0 < t_1 < \dots < t_N = T$ is a regular grid of observation dates, i.e. $t_k = kT/N$ for $k = 0, \dots, N$.

- ▶ Som examples
 - ▶ Call option : $N = 1$ and $\varphi(S_0, S_1) = (S_1 - K)_+$.
 - ▶ Discrete barrier option : $T = 1, N = 12$ and $\varphi(S_0, S_1, \dots, S_{12}) = (S_{12} - K)_+ \mathbf{1}_{\{\forall 0 \leq i \leq 12, S_i \geq L\}}$.

Objectives (2)

- Build a portfolio V such that

$$V_T = \varphi(S_{t_0}, S_{t_1}, \dots, S_{t_N})$$

$$V_t = H_t^0 S_t^0 + H_t \cdot S_t$$

$$dV_t = H_t^0 dS_t^0 + H_t \cdot dS_t$$

where H^0 is the quantity of non-risky asset $S_t^0 = e^{rt}$ and H is the quantity of risky assets.

- In continuous time, such a V exists thanks to the martingale representation theorem

$$V_t = e^{-r(T-t)} \mathbb{E}[\varphi(S_{t_0}, S_{t_1}, \dots, S_{t_N}) | \mathcal{F}_t] := u(t, S_t)$$

$$H_t = \partial_d u(t, S_t).$$

- Design and implement a tool in python to build a discrete approximation of V .

Requirements

- ▶ Approximate expectations
 - ▶ Use empirical average instead of approximations

$$\mathbb{E}[\varphi(S_{t_0}, S_{t_1}, \dots, S_{t_N})] \approx \frac{1}{M} \sum_{m=1}^M \varphi(S_{t_0}^{(m)}, S_{t_1}^{(m)}, \dots, S_{t_N}^{(m)})$$

- ▶ Always provide a confidence interval

$$\xi_M^2 = \left[\frac{1}{M} \sum_{m=1}^M (\varphi(S_{t_0}^{(m)}, \dots, S_{t_N}^{(m)}))^2 - \left(\frac{1}{M} \sum_{m=1}^M \varphi(S_{t_0}^{(m)}, \dots, S_{t_N}^{(m)}) \right)^2 \right]$$

$$95\% \text{ CI} : \left[\frac{1}{M} \sum_{m=1}^M \varphi(S_{t_0}^{(m)}, \dots, S_{t_N}^{(m)}) \pm \frac{1.96 \xi_M}{\sqrt{M}} \right].$$

- ▶ Use finite difference methods to approximate partial derivatives.

The multi-dimensional Black-Scholes model

$$S_{t,d} = S_{0,d} e^{(r - (\sigma_d)^2/2)t + \sigma_d B_{t,d}}, \quad d = 1, \dots, D$$

where σ_d are real valued volatilities, B_d are real valued Brownian motion for all $1 \leq d \leq D$ and $\frac{\text{Cov}(B_t, B_t)}{t} = \Gamma$.

There exists L such that $LL' = \Gamma$. Then,

$$S_{t,d} = S_{0,d} e^{(r - (\sigma_d)^2/2)t + \sigma_d L_d W_t}, \quad d = 1, \dots, D$$

where L_d denotes the d^{th} -row of the matrix L and $W = (W_1, \dots, W_D)$ is Brownian motion with values in $\mathbb{R}^D$. The components of W are independent.

Sampling the Black-Scholes model

- Note that for all $t, u \geq 0$

$$S_{t+u,d} = S_{t,d}^d e^{(r-\sigma^2/2)u + \sigma(B_{t+u,d} - B_{t,d})} = S_t \tilde{S}_u$$

- Consider a regular grid $t_k = \frac{kT}{H}$ for $k = 0, \dots, H$

$$S_{t_{i+1},d} \stackrel{Law}{=} S_{t_i,d} e^{(r-(\sigma_d)^2/2)\frac{T}{H} + \sigma_d \sqrt{\frac{T}{H}} L_d G_{i+1}}, \quad d = 1, \dots, D$$

where $(G_i)_{i \geq 1}$ is iid $\mathcal{N}(0, I)$.

- Let $t_i \leq t < t_{i+1}$, we consider the problem of sampling $(S_{t+u}, u \geq 0)$ knowing $(S_s, s \leq t)$. For $d = 1, \dots, D$,

$$S_{t_{i+1},d} \stackrel{Law}{=} S_{t,d} e^{(r-(\sigma_d)^2/2)(t_{i+1}-t) + \sigma_d \sqrt{(t_{i+1}-t)} L_d G_{i+1}}$$
$$S_{t_{k+1},d} \stackrel{Law}{=} S_{t_k,d} e^{(r-(\sigma_d)^2/2)\frac{T}{H} + \sigma_d \sqrt{\frac{T}{H}} L_d G_{k+1}}, \quad k = i+1, \dots, D$$

The discrete time portfolio

- ▶ Consider a regular time grid $0 = \tau_0 < \dots < \tau_H = T$ with step size T/H .
- ▶ Let $\hat{p}_0 \approx u(0, S_0)$ and $\delta_i \approx \nabla u(\tau_i, S_{\tau_i})$.
- ▶ The value of the discrete time hedging portfolio is given by

$$V_i = \delta_i \cdot S_{\tau_i} + \phi_i^0 S_{\tau_i}^0$$

where ϕ_i^0 is determined using the self-financing equation

$$(V_{i+1} - V_i) = \delta_i \cdot (S_{\tau_{i+1}} - S_{\tau_i}) + \phi_i^0 (S_{\tau_{i+1}}^0 - S_{\tau_i}^0)$$

- ▶ Let us introduce $\tilde{V}_i = \frac{V_i}{S_{\tau_i}^0}$. Then,

$$(\tilde{V}_{i+1} - \tilde{V}_i) = \delta_i \cdot (\tilde{S}_{\tau_{i+1}} - \tilde{S}_{\tau_i}).$$

- ▶ The hedging error at maturity time is given by
 $P\&L = p_0 + e^{r\tau_H} \tilde{V}_H - \text{payoff}.$

Options to implement

- ▶ Basket option (key basket)

$$\left(\sum_{d=1}^D \lambda_d S_{T,d} - K \right)_+$$

where $\lambda_i \in \mathbb{R}$ and $K \in \mathbb{R}$. The case $K \leq 0$ enables us to consider a put like option.

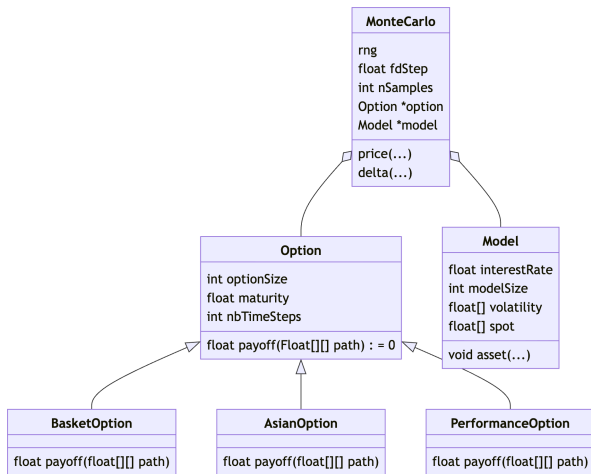
- ▶ Discrete Asian option (key asian)

$$\left(\sum_{d=1}^D \lambda_d \frac{1}{N+1} \sum_{i=0}^N S_{t_i,d} - K \right)_+.$$

- ▶ Performance option on a basket (key performance)

$$1 + \sum_{i=1}^N \left(\frac{\sum_{d=1}^D \lambda_d S_{t_i,d}}{\sum_{d=1}^D \lambda_d S_{t_{i-1},d}} - 1 \right)_+.$$

Design



Good practice

- ▶ Write tests ! Tests are part of the code.
- ▶ Carefully test every function.
- ▶ Document your code.
- ▶ Use the vanilla call option
 - ▶ to check your Monte-Carlo implementation,
 - ▶ to check your finite difference approximation,
 - ▶ to check your portfolio equation.